

Jack Balster

(206) 966-1791 | jack.balster@wsu.edu | <https://www.linkedin.com/in/jack-balster/> | <https://github.com/jack-balster>

EDUCATION

Washington State University

- M.S. Software Engineering
- B.A. Management Information Systems
- Minor Computer Science

Expected Graduation: May 2026

GPA: 3.8

President's Honor Roll: 4 years

Distinguished Regents Scholar Award Winner

PROJECTS

WSU Tutoring Center Portal

2023

Enhanced and redesigned tutoring portal with expanded features

Python | Django | JavaScript | MySQL | Bootstrap

- Developed an enhanced tutoring system for WSU, expanding upon the existing portal with additional features and a comprehensive redesign of the user interface.
- Implemented a multi-role user system which includes students, tutors, and admins each with different access privileges and functionalities within the portal.
- Integrated features such as scheduling appointments based on availability, reviewing tutors, cancelling and viewing appointments, filtering by major, specific classes, or by tutor, creating majors and classes, edit profile, and more.
- Incorporated advanced features such as confirmation emails, password recovery, and extensive tutor reports using ReportLab for statistical analysis of tutors and tutoring sessions.
- **GitHub:** <https://github.com/jack-balster/Tutoring-Portal>

Real-time Chat Platform

2024

Chat app comparable to telegram and iMessage

JavaScript | React | Node.js | Express.js | MongoDB | TailwindCSS | Firebase

- Developed a comprehensive chat platform to enable efficient and interactive communication between users.
- Implemented user authentication with Firebase and built a responsive front-end using React and TailwindCSS.
- Enabled real-time communication with Socket.io, allowing users to send messages instantaneously across devices.
- Utilized MongoDB for scalable data storage, management of user profiles, chat histories, and application settings.
- Included key functionalities such as user registration, profile customization, chat room creation, and real-time status updates such as active user indicators, message timestamps, and read receipts.
- **GitHub:** <https://github.com/jack-balster/Real-Time-Chat-Platform>

Yelp Business Review Analysis Application

2024

Business insights application leveraging Yelp data for enhanced analysis

Python | PostgreSQL | Qt Designer | Lucidchart | SQL

- Employed Lucidchart for the initial design and visualization of the database schema, ensuring an optimal structure.
- Utilized advanced SQL techniques for data manipulation and to implement complex queries such as filtering by state, city, zip code, and category. Also included custom metrics to judge the success of businesses.
- Developed and maintained a PostgreSQL database, incorporating SQL queries to extract, analyze, and present actionable business data through a Qt Designer-built UI.
- **GitHub:** <https://github.com/jack-balster/Yelp-Business-Review-Analysis>

Facial Recognition Interface

2024

Multi-functional facial recognition system with analytic capabilities

Python | C++ | SQLite | OpenCV | TensorFlow | Docker

- Developed a facial recognition system using advanced machine learning frameworks like TensorFlow and computer vision libraries such as OpenCV.
- Integrated functionalities such as user recognition, emotion detection, gender prediction, and age estimation.
- Designed an intuitive graphical interface with Tkinter, allowing users to create profiles, train the model, and utilize facial recognition features efficiently.
- Containerized the application with Docker, facilitating deployment and scalability across various platforms.
- **GitHub:** <https://github.com/jack-balster/Facial-Recognition-Interface>

TECHNICAL SKILLS

Languages | C, C++, C#, Python, JavaScript, VB.NET, HTML, CSS, SQL

Frameworks / Libraries | ASP.NET, Plotly, Tailwind, Flask, React, Node.js, Express.js, Django, Scikit-Learn, Pandas, OpenCV

Databases / Tools | PostgreSQL, SQLite, MongoDB, Microsoft SQL Server, Git, Docker, Tableau, Lucidchart